| | | | |
|---|---|---|---|
| **Fiscal Year:** | FY 2017 | **Task Last Updated:** | FY 04/04/2017 |
| **PI Name:** | Ferl, Robert Ph.D. | | |
| **Project Title:** | Hypobaric Plant Biology - Molecular Responses of Arabidopsis to the Low Atmospheric Pressures of Spaceflight Vehicles and Planetary Habitats | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Plant Biology | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | (1) Bioregenerative Life Support | | |
| **PI Email:** | robferl@ufl.edu | **Fax:** | FY 352-392-4072 |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 352-392-1928 |
| **Organization Name:** | University of Florida | | |
| **PI Address 1:** | Horticultural Sciences, 1301 Fifield Hall | | |
| **PI Address 2:** | Box 110690 | | |
| **PI Web Page:** | http://www.hos.ufl.edu/ | | |
| **City:** | Gainesville | **State:** | FL |
| **Zip Code:** | 32611-0690 | **Congressional District:** | 3 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2012 Space Biology NNH12ZTT001N |
| **Start Date:** | 07/02/2013 | **End Date:** | 12/31/2016 |
| **No. of Post Docs:** | 2 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA KSC |
| **Contact Monitor:** | Levine, Howard | **Contact Phone:** | 321-861-3502 |
| **Contact Email:** | howard.g.levine@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 12/31/2016 per NSSC information (Ed., 5/6/16) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Paul, Anna-Lisa Ph.D. ( Co-PI: University of Florida ) | | |
| **Grant/Contract No.:** | NNX13AM46G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

| | |
|---|---|
| **Task Description:** | Atmospheric pressure and composition are among the engineering variables considered in the design and construction of spaceflight vehicles and extraterrestrial habitats. Simply put, the costs of maintaining a pressure vessel at one atmosphere have been traded away throughout the history of spaceflight vehicle design and are traded away in future designs, yet the biological impacts of lower atmospheric pressure, especially on plants, is both profound and poorly understood. The objective of this proposal is to develop a refined understanding of the metabolic processes involved in plant responses and physiological adaptations to low pressure environments relevant to space exploration. The long-term goal of this line of research is a fundamental understanding of low pressure plant biology within exploration vehicles and structures, with a practical goal of producing plants that are specifically designed to thrive in low atmospheric pressures. The essential drivers of this project are the inescapable engineering limitations to producing orbital, lunar, or Martian plant growth facilities that contain Earth-normal atmospheric pressures, and the knowledge that plants do mount complex, and sometimes unexpected costly metabolic responses to hypobaria. These responses can also be used as a model that will allow fundamental understanding of how plants adapt to changing terrestrial environments. In the initial phases of the project, we will use the tools of functional genomics in Arabidopsis plants to examine molecular genetic responses that occur during initial and chronic exposure to atmospheres ranging from Earth normal to near Martian pressures and containing various gas mixtures, especially those atmospheres that are in future vehicle designs. We will build on our previous molecular work using Arabidopsis (Paul et al., 2004). We will conduct extensive tests of a range of appropriate pressures and compositions using wild type genotypes. Then we will use select mutants and engineered changes in metabolic pathways implicated by the initial gene expression array data to test hypotheses regarding the fundamental mechanisms associated with the hypobaria response. Finally, we proposed a novel use of International Space Station (ISS) airlock facilities to investigate low pressure processes directly in a spaceflight microgravity environment, to take a first look at the emergent effects of altering atmospheric pressure within an operational vehicle in space; however, this last aspect of the proposal was not funded.<br>The overall goal of our program is to understand fundamental molecular biological responses to extraterrestrial and spaceflight environments. The current proposal for ground-based research leading to spaceflight addresses specifically a focus area defined by the Decadal Study: Plant and Microbial Growth Under Altered Atmospheric Pressures. The proposed research will provide fundamental insights into the biological impact of novel atmospheric environments, a focus area that is specifically identified in the Decadal Study (Decadal-Survey-Committee 2011).<br><br>Reference:<br><br>Paul AL, Schuerger AC, Popp MP, Richards JT, Manak MS, and Ferl RJ. (2004) Hypobaric biology: Arabidopsis gene expression at low atmospheric pressure. Plant physiology 134:215-223. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Experiments in extreme and novel environments, like our spaceflight program and these in low pressure habitats, tell us that that life utilizes special, potentially unique signals to adapt to living with unique challenges. This insight has tremendous implications for the expansion of human existence, beyond, to other worlds, but also richly informs us about the potential for plants to adapt to environmental changes here on Earth. Any insights we reveal have applications to understanding how plants may better cope with some of the stresses we see in our daily environment like drought, global warming, flooding, high CO2, etc. No plant on Earth has ever evolved to respond to ultra-low atmospheric pressures, yet with a little help, plants do just fine in this environment. We are learning what physiological tools plants use to cope with novel stresses, and that teaches us two things. First, it helps us understand what kind of help plants need to be productive on a long spaceflight mission or in a greenhouse on Mars--not an immediate need, but information we need before we do either of those things. Second, how a plant responds to a novel stress gives us a fundamental understanding of how plants cope with an adverse environment. This knowledge can help us predict how plants of the Earth will manage changes in climates that we have not yet seen on a global scale, or how plants might respond to new pollutants.<br>This research is also about understanding how to optimize protected habitats and closed environments, issues that are important to us if we want to grow plants on a space station or long journey, but also if we want to grow plants in a greenhouse on another planetary surface. The lessons we have learned and the tools we have developed growing plants in low pressure chambers, as well as a Martian-analog greenhouse in the Arctic, have direct application to protected agriculture in Florida, and anywhere on Earth. |
| | The summaries of a variety of experiments that comprised this body of work are provided below.<br>I. Overview of Plant Transcriptome Response to Different Levels of Low Pressure.<br><br>Two experiments were used to establish an overview of molecular responses through the use of Affymetrix ATH1 arrays for roots and for shoots. Unless noted otherwise, in all cases gene expression showing at least 2-fold change in abundance and $p < 0.01$ were defined as differentially expressed genes:<br><br>Experiment 1 – Variations on a Pressure. A set of 10 days old plants grown vertically on plates were transferred to the LPGCs and exposed to six different atmospheric pressures for a period of 24 hours; 97 kPa, 75 kPa, 50 kPa, 25 kPa, 10 kPa, and 5 kPa. In all treatments the carbon dioxide was kept consistent at a partial pressure of 0.05 kPa. In each treatment, oxygen was kept at 21% of the total chamber pressure with Nitrogen as a balance of remaining gas. Each treatment of atmospheric pressures and gases was replicated with three separate, concurrently run LPGC. Each chamber in the LPGCs contained 10 plates, with approximately 12 plants on each plate.<br><br>A total of 3156 genes were differentially expressed by at least 2-fold in at least one hypobaric treatment, and yet not a single gene was co-ordinately expressed among all of the treatments in leaves, and only three genes showed coordinate expression in roots. For treatments at pressures between 75 kPa and 10 kPa the transcriptomes showed dramatic tissue-specific patterns; roots shared few differentially expressed genes in common with shoots in response to hypobaria. In addition, roots appeared more sensitive to these environmental changes, as far more genes were significantly changed in roots than in shoots. Most of the genes that are repressed in hypobaric environments are associated with growth and metabolism. In shoots, many of the differentially expressed genes at 50 kPa are associated with transport functions and pathogen response rather than hypoxia. Below 50 kPa, the induction of genes associated with desiccation and water deprivation responsive pathways predominated.<br><br>Experiment 2 – A 10 kPa Time Course. A set of 10 day old plants grown vertically on plates were exposed to 10 kPa |

and 97 kPa in LPGC and sampled at six different time points. The 97 kPa atmosphere was composed of a partial pressures of 21 kPa oxygen, 0.05 kPa carbon dioxide, and a balance of nitrogen. The 10 kPa samples were composed of a partial pressure of 2.1 kPa oxygen, 0.05 kPa carbon dioxide, and a balance of nitrogen. The samples were harvested at: 1 hour, 3 hours, 6 hours, 12 hours, 48 hours, and 72 hours. Each treatment of atmospheric pressures and times were replicated with three separate, concurrently run LPGC. Each chamber in the LPGCs contained 10 plates of approximately 12 plants each.

There were a total of 2801 genes that were differentially expressed in at least one time point over the course of the 3 day experiment. Only 200 of those genes (7%) were coordinately expressed with similar trends in both roots and shoots. The consistently up-regulated genes included typical hypoxia responsive genes as wells as drought and cold marker genes.

Roots respond more quickly than shoots, as demonstrated by the large number of induced genes at 1 h. In contrast, the intensity of the response appears to generally mount over time in shoots, particularly with respect to down-regulated genes. The early response in roots could be promoted by the hypoxic component of hypobaria; the plant low-oxygen response consists of approximately two stages in a previous time course study, 0 to 0.5 h and 2 to 20 h. Shoots exhibited fewer differentially expressed genes in the early time points than roots. Compared to roots, shoots also have fewer examples of genes which display a mounting degree of expression over time. Nonetheless, there were many genes associated with biosynthetic and metabolic process that were induced or repressed early, and maintained or exceeded those levels of expression over time.

CONCLUSIONS for Overview of Plant Transcriptome Response to Different Levels of Low Pressure.

The unique differential gene expression patterns in response to hypobaria suggest that the physiological adaptation of Arabidopsis to reduced atmospheric pressure is more complex than the acclimation to the reduced partial pressures of oxygen inherent to low atmospheric pressures. Arabidopsis plants respond to hypobaria with a variety of changes in gene expression patterns that are organ specific, change with respect to the severity of the hypobaric environment, and adjust over time. The hypobaric response appears to overlay the perception of desiccation and biotic challenge on top of the perceptions of hypoxic stress, which suggests that the practical compensation for hypobaric stress in plants cannot be as simple as increasing the oxygen content, as is done for humans in the low pressure environments of space vehicles and space suits, or on the top of Mt. Everest.

These studies provide a window into the nature of the strategies used for physiological adaptation to hypobaria as an example of novel environments, and can allow dissection of specific components of plant responses to hypobaria as part of space exploration mission habitat design. For example, in a well hydrated, yet hypobaric greenhouse on Mars, plants would not need to expend the metabolic energy to maintain the drought response the hypobaria would elicit. Thus, there could be an advantage to engineer the plants along with the habitats envisioned for planetary exploration. The more fundamental application of this physical phenomenon is that plant metabolic responses to drought stress and stomatal regulation can be examined without subjecting the plant to comprehensive desiccation.

In addition, these transcriptome analyses may reveal genes that have contributed to the natural adaptation of plants to high altitude environments. It has been argued that plants that have evolved to live in a “stressful” environment are not actually “stressed” at all, rather, this is the habitat that is normal to them, and so the metabolic processes employed are just as benign as those engaged by their cousins living in seemingly more hospitable climates. Indeed, many of the differentially expressed genes at 75 kPa reflect gene expression patterns that are also differentially regulated in high altitude species and ecotypes when compared to representatives indigenous to lower altitudes.

As a unique stress, hypobaria presents an opportunity to understand the development of plant responses to novel environmental stress. We conclude that some aspects of the overall response to hypobaria are appropriate, in that hypoxia responses to the low oxygen of hypobaria make sense. However we also conclude that some elements of the response are inappropriate, in that the drought responses that are seen in hypobaria are inconsistent with a wet and humid environment. These seemingly inappropriate responses may help illuminate drought perception mechanism in plants, and further suggest strategies for improving growth in hypobaria especially as pressure is considered in extraterrestrial habitats and vehicles.

II. Dissection of Hypobaric Response: Hypoxia and Hypobaria with Normoxia

Controlled hypobaria presents biology with an environment that is never encountered in terrestrial ecology, yet the apparent components of hypobaria are stresses typical of terrestrial ecosystems. High altitude, for example, presents terrestrial hypobaria always with hypoxia as a component stress, since the relative partial pressure of O2 is constant in the atmosphere. Laboratory-controlled hypobaria, however, allows the dissection of pressure effects away from the effects typically associated with altitude, in particular hypoxia, as the partial pressure of O2 can be varied. In this study, whole transcriptomes of plants grown in ambient (97 kPa/pO2=21 kPa) atmospheric conditions were compared to those of plants transferred to five different atmospheres of varying pressure and oxygen composition for 24 hours: 50 kPa/pO2=10 kPa, 25 kPa/pO2=5 kPa, 50 kPa/pO2=21 kPa, 25 kPa/pO2=21 kPa, or 97 kPa/pO2=5 kPa. The plants exposed to these environments were 10 day old Arabidopsis seedlings grown vertically on hydrated nutrient plates. In addition, 5 day old plants were also exposed for 24 hrs to the 50 kPa and ambient environments to evaluate age-dependent responses. The gene expression profiles from roots and shoots showed that the hypobaric response was more complex than simple hypoxia, and that adding back oxygen to normoxic conditions did not completely alleviate hypobaric responses.

Plant phenotypes in low atmospheric pressure environment-- There was no evidence of desiccation-associated phenotype such as leaf wilting observed in either the 10 day old (10 d) or 5 day old (5 d) seedlings across the hypobaric and hypoxic treatments.The high relative humidity and continuous water supply provided by the micro-climate of the Petri plate growing system likely removes any outward expression of actual dehydration. The solid nutrient media in the petri plant growing system is sufficient to grow Arabidopsis for 24 hours in severe hypobaria without exhausting the water supply within the media. Although water is moving more rapidly through the stomata, this phenomenon does not cause a visible (e.g., wilted) phenotype.

The impact of supplemental oxygen on transcriptional responses to 50kPa atmospheres. Two treatments were used to evaluate the contribution of hypoxia to the hypobaric transcriptome of Arabidopsis: exposing seedlings to 50 kPa, and to 50 kPa supplemented with oxygen to a normal (21 kPa) level (50 kPa/NormOx). Genes with significant ($p < 0.01$) differential expression by at least 2-fold based on 97 kPa control were defined as differentially expressed genes. There

**Task Progress:**

were 151 differentially expressed genes identified in at least one of these two conditions in roots or shoots, which were defined as 50 kPa atmosphere associated genes of 10d plants. Most of these genes exhibited different expression patterns in roots and shoots, demonstrating the tissue-specific response to hypobaria. Among them, 112 genes in 50 kPa and 25 genes in 50 kPa/NormOx were significantly changed in roots, while 23 genes in 50 kPa and 2 genes in 50 kPa/NormOx were in shoots

The impact of developmental age on the transcriptional response to 50 kPa atmospheres-- The developmental age of Arabidopsis had a substantial effect on the profile of differentially expressed genes in response to the 50 kPa and 50kPa/NormOx atmospheres. The root transcriptomes of 5 d Arabidopsis seedlings revealed exhibited notably different expression patterns than the transcriptomes of the 10 d roots in the same environments. There were 221 differentially expressed genes in at least one of these two conditions in roots of 5 d or 10 d plants, which were subsequently defined as 50 kPa atmosphere associated genes of 5 d/10 d roots. Among them, 103 genes of 5 d roots and 129 genes of 10 d roots were identified. Only 10 genes were co-ordinately expressed in 5 d and 10 d roots in 50 kPa, and just a single gene in the 50kPa/NormOx treatments.

Comparison of transcriptional responses to hypobaria and hypoxia with respect to 25 kPa atmospheres-- The more severe hypobaric environment of 25 kPa was used to further dissect the contribution of oxygen in the transcriptional response of 10 d Arabidopsis to reduced atmospheric pressure. The partial pressure of oxygen at 25 kPa is 5 kPa, an oxygen concentration which is known to confer a strong hypoxic stress response in plants. To better dissect hypobaric stress in 25 kPa, two additional conditions were utilized: total gas pressure in 25 kPa with a partial pressure of supplemental oxygen in 21 kPa (defined as 25kPa/NormOx) and hypoxia containing total gas pressure in 97 kPa with a partial pressure of oxygen in 5 kPa (defined as 97kPa/HypOx). Array analysis showed that there were 372 differentially expressed genes in at least one condition of 25 kPa, 25kPa/NormOx or 97kPa/HypOx in roots or shoots, which were therefore defined as 25 kPa atmosphere associated genes of 10 d plants.

Similar to 50 kPa and 50kPa/NormOx, difference of transcription pattern between responses to 25 kPa and 25kPa/NormOx as well as 25 kPa and 97kPa/HypOx must be clarified through direct comparison. Among 372 of 25 kPa atmosphere associated genes identified in as differentially expressed genes in response to 25 kPa, 25kPa/NormOx and 97kPa/HypOx in roots and shoots of 10 d plants, there were 271 genes in roots and 56 genes in shoots significantly changed in at least one of 25 kPa and 25kPa/NormOx, and 298 genes in roots and 74 genes in shoots significantly changed in at least one of 25 kPa or 97kPa/HypOx. In total, 93 genes between 25 kPa and 25kPa/NormOx and 5 genes between 25 kPa and 97kPa/HypOx showed more than 1.8-fold in expression change in roots, and 44 genes between 25 kPa and 25kPa/NormOx and 11 genes between 25 kPa and 97kPa/HypOx detected with same criteria in shoots. These genes represented the differential response to “hypobaria together with hypoxia” (25 kPa) and “hypobaria without normoxia” (25kPa/NormOx) or “hypoxia” (97kPa/HypOx). For severe low atmospheric pressure level (25 kPa) in which hypoxic stress was strong, the difference between responses to 25 kPa and 97kPa/HypOx were quite small in both roots and shoots. Majority of genes differentially expressed between responses to 25 kPa and 25kPa/NormOx did not show difference between responses to 25 kPa and 97kPa/HypOx. However, most of genes differentially expressed between 25 kPa and 97kPa/HypOx were also significantly changed between 25 kPa and 25kPa/NormOx, suggesting that a large portion of hypobaric responses were prompted by hypoxia, not total gas pressure, in this oxygen level.

Adjustment of oxygen in space missions-- Being able to mitigate the effects of atmospheres modified from “Earth normal” is central to the exploration mission to take biology off planet, but understanding how plants dissect components of their environment to physiologically adapt to a stress is crucial to all plant cultivation scenarios. In the dissection of the hypobaria and hypoxic responses, the organ-specific and age-dependent transcription patterns suggest plants use adaptive strategies tied to the unique needs of a specific tissue, and that those needs change as the plant develops. Notably, the continuous light that matched the hardware in International Space Station (ISS) was used in our experimental process including plant germination, growth, and all treatments. It was reported that light signal could affect hypoxic response especially in tissue-dependent gene induction. Low oxygen promotes accumulation of diverse metabolites in roots and shoots as well as in plants treated with different levels (5-10 kPa or ~ 0 kPa) or types (submergence, flooding or argon-gassed) of hypoxia. Consistent with previous study of hypoxia response, 97kPa/HypOx (97 kPa/pO2=5 kPa) highly induced ethanol metabolic genes including ADH1 (AT1G77120) and PDC2 (AT5G54960) as well as alanine associated gene ALAAT1 (AT1G17290) in roots but not shoots. However, in a recent study most of the previously defined core hypoxia genes (Mustroph et al., 2009) were induced in an organ-independent manner by flooding when plants were submerged in early stage of photoperiod, while in our present data 34 out of 51 core genes were associated with 25 kPa atmosphere and only 9 genes showed similar induction by 97kPa/HypOx in roots and shoots. These suggest that the growth condition with constant light started from germination might alter organ-specific regulation of hypoxia genes compared with other photoperiod conditions. Considering the facility limitation in spaceflight mission, it will be necessary to conduct respective studies on plant responses to oxygen changes in allusion to different light conditions of growth. The oxygen supplement assays provided novel insight into plant transcriptomic response in the condition of low atmospheric pressure with adjusted partial pressures of O2, such as has been utilized in manned space missions in the past, and may be employed in the future. When oxygen is supplemented to both moderate and severe hypobaric conditions it minimizes transcriptional responses caused by low atmospheric pressure, with the implication that the overall stress load on the plant is reduced. In comparison between hypobaria without hypoxia in 50 kPa and 25 kPa, 40 genes in roots and 3 genes in shoots were significantly changed in at least one of 50kPa/NormOx and 25kPa/NormOx conditions. The normoxic conditions with different total gas pressure resulted in similar transcription trends in these genes from both roots and shoots, which supported the conclusion from comparisons of 50 kPa vs 50kPa/NormOx and 25 kPa vs 25kPa/NormOx that plants were not so sensitive to hypobaria with nomoxia compared with hypobaria with hypoxia. However, while increasing oxygen concentrations in hypobaric environments mediates stress in plants, the data presented here suggest that exposing plants to at least mild hypobaric conditions can also serve to mediate plant stress in low oxygen environments. Interestingly, although the importance of oxygen in human hypobaric stress, such as the high-altitude pulmonary edema common to extreme mountain climbers, has long been appreciated, it is becoming increasingly clear that the combination of stresses associated with hypobaric environments elicits complex physiological responses, and requires complex mediation. The establishment of protected agriculture in extreme environments -- whether in on the surface of Mars, in exploration vehicles, or for terrestrial applications – will require careful management of in situ resources and engineering considerations to optimize the internal environment for plant health and productivity. For instance, higher gas pressure differentials between inside and outside of greenhouse structures significantly increases the leak rate and therefore any reduction of internal pressure will bring engineering benefits. Further, the ability to lower specific components of the gas mixture to grow plants would bring a benefit of reduced transportation of gases and increased payload. However, regardless of the engineering benefits, it is imperative that the biological consequences of hypobaric environments be well understood before adopting

| | |
|---|---|
| | this strategy for exploration life support habitats.<br><br>Conclusion<br><br>The data presented here show that plant physiological adaptation to hypobaria is certainly more involved than a simple response to hypoxia. Based on the patterns of gene expression in hypobaria with supplemental oxygen, the most influential feature of hypobaria beyond hypoxia is a water stress apparently associated with the overall reduction of gas pressure, which manifests as mimicked aspects of drought stress. However, gene expression patterns also suggest that a plant's response to hypobaria is more complex than simply a combination of hypoxia and drought stress. The gene expression patterns of plants in a normoxic low pressure environment suggest that plants are engaging genes associated with a complex stimulus that induces a novel metabolic pattern involving multiple unknown signaling components that is distinct from effects derived from exposure to either hypoxia or drought. Further, these data revealed that the molecular responses are organ specific in that roots and shoots each possess distinct metabolic strategies for adjusting to hypobaria. Taken together these data suggest that while the stress response to the novel environment of hypobaria can be largely understood as an appropriate response to the component stresses of hypoxia and water loss, those two components do not fully explain responses to hypobaria.<br><br>The data presented here are still under analysis, and ongoing work for publication will continue.<br><br>Reference<br><br>Mustroph A, Zanetti ME, Jang CJ, Holtan HE, Repetti PP, Galbraith DW, Girke T, Bailey-Serres J. Profiling translatomes of discrete cell populations resolves altered cellular priorities during hypoxia in Arabidopsis. Proc Natl Acad Sci U S A. 2009 Nov 3;106(44):18843-8. doi: 10.1073/pnas.0906131106. |
| **Bibliography Type:** | Description: (Last Updated: 11/27/2022) |
| **Articles in Peer-reviewed Journals** | Zhou M, Callaham JB, Reyes M, Stasiak M, Riva A, Zupanska AK, Dixon MA, Paul A-L, Ferl RJ. "Dissecting low atmospheric pressure stress: transcriptome responses to the components of hypobaria in Arabidopsis." Frontiers in Plant Science. 2017 Apr;8:528. http://dx.doi.org/10.3389/fpls.2017.00528 ; PubMed PMID: 28443120; PubMed Central PMCID: PMC5385376 , Apr-2017 |
| **Articles in Peer-reviewed Journals** | Paul A-L, Zhou M, Callaham JB, Reyes M, Stasiak M, Riva A, Zupanska AK, Dixon MA, Ferl RJ. "Patterns of Arabidopsis gene expression in the face of hypobaric stress." Annals of Botany (AoB Plants) - In revision, as of April 2017. Expected publication September 2017. , Apr-2017 |
| **Articles in Peer-reviewed Journals** | Zhou M, Paul A-L, Ferl RJ. "Data for characterization of SALK_084889, a T-DNA insertion line of Arabidopsis thaliana." Data Brief. 2017 May 31;13:253-8. eCollection 2017 Aug. https://doi.org/10.1016/j.dib.2017.05.047 ; PubMed PMID: 28649584; PubMed Central PMCID: PMC5470433 , May-2017 |